

Report: Coffee Sales Data Analysis and Prediction

- **Objective**

The aim of this project was to analyze and model a dataset containing coffee sales data. Key objectives included:

Cleaning and preprocessing the data to handle missing values, outliers, and inconsistencies.

Conducting Exploratory Data Analysis (EDA) to uncover trends in sales.

Developing a linear regression model to predict sales (money) based on features.

- **Implementation Steps**

1. Data Loading

The dataset was loaded from the specified file path using pandas.

The first few rows were printed to understand the structure of the data.

2. Handling Missing Values

Numerical Columns: Missing values in the money column (if present) were filled with the column's median. Median was chosen to handle skewness in the data.

Categorical Columns: Missing values in the coffee_name column (if present) were replaced with the mode (most frequent value).

Datetime Conversion: The date column was converted to datetime format, handling errors gracefully by coercing invalid entries.

3. Outlier Removal

Z-scores were calculated for the money column to identify and exclude extreme outliers (Z-score threshold = 3). This ensured the model was trained on reliable data without noise caused by extreme values.

4. Feature Engineering

The date column was leveraged to extract new features: Month and Year. These features provided temporal insights into sales trends.

5. Exploratory Data Analysis (EDA)

Monthly Sales Trends: A line plot was created to visualize sales (money) trends across months, with separate lines for different years.

Sales by Coffee Type: A bar plot was used to analyze sales performance by coffee type.

6. Machine Learning

Preprocessing:

The target variable (money) was separated from the features.

Categorical features were encoded using one-hot encoding to prepare the data for modeling.

Splitting the Data:

Data was split into training and testing sets (80% training, 20% testing).

Model Development:

A linear regression model was trained on the training dataset.

Predictions were made on the test dataset.

Model Evaluation:

The model's performance was evaluated using Mean Squared Error (MSE) and R-squared (R^2) metrics.

Coefficients of the model were displayed for interpretability.

• **Problem-Solving Approach**

Handling Missing Data:

Replacing missing numerical and categorical values with median and mode respectively ensured data integrity without introducing biases.

Outlier Removal:

Z-score methodology effectively removed outliers, reducing noise in the dataset.

Feature Extraction:

Extracting Month and Year from the date column enriched the dataset for temporal analysis.

Data Splitting:

The use of train-test split ensured an unbiased evaluation of the model's performance on unseen data.

- **Challenges and Solutions**

Missing Columns:

During development, there was a risk of missing expected columns like money, coffee_name, or date.

Solution: Conditional checks (if column in data.columns) were implemented to avoid runtime errors.

Outlier Handling:

Outliers in money could significantly skew the model.

Solution: Used Z-score with a threshold to identify and remove extreme values.

Datetime Conversion Errors:

Inconsistent or invalid date formats could cause conversion failures.

Solution: Used errors='coerce' in pd.to_datetime to handle invalid dates gracefully.

Categorical Encoding:

Handling categorical variables required careful encoding to avoid issues like the dummy variable trap.

Solution: Used pd.get_dummies with drop_first=True to perform one-hot encoding efficiently.

Multicollinearity in Features:

Features derived from Month and Year may introduce multicollinearity, leading to biased coefficients.

Solution: Ensured proper encoding and feature selection to mitigate this risk.

- **Results**

Model Performance:

The Mean Squared Error (MSE) provided an estimate of prediction errors.

The R^2 score indicated how well the model explained the variability in sales data.

Visual Insights:

Sales trends over time and performance by coffee type were visualized, providing actionable insights for decision-making.

- **Conclusion**

This project successfully demonstrated the ability to clean, analyze, and model coffee sales data. By handling missing values, outliers, and feature engineering, the dataset was prepared for effective modeling. The linear regression model provided a baseline for predicting sales, which could be further improved with advanced algorithms.